CareQ

An Integrated Healthcare Mobile Application for Malaysia

Proposal submitted by:

No	Student Name	Matric No
1	Kasem Bau	MBS244044
2	Kemi Yonaniko	MBS244050
3	Hardev Singh A/L Harpal Singh	MBS244056
4	Nor Shahira Binti Md Isa	MBS244062
5	Noraisyah Binti Nordin	MBS244082

2026 · Healthcare Technology Proposal

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1. Executive Summary

CareQ is a proposed integrated healthcare mobile application designed to address the persistent challenges faced by patients and healthcare providers in Malaysia. The application aims to bridge the gap between patients seeking medical care and the clinics providing it, using digital tools to reduce waiting times, improve transparency, and empower individuals to take charge of their own health.

The Malaysian healthcare system, while comprehensive, continues to struggle with overcrowding, long queues, and a lack of patient-facing digital infrastructure, particularly in government clinics. CareQ proposes a unified platform that combines appointment booking, real-time queue tracking, health self-monitoring, AI-assisted symptom checking, and lifestyle guidance into a single, accessible mobile application.

This proposal outlines the full scope of the application. Its features, user journey, technical approach, expected impact, and implementation roadmap. The goal is to demonstrate how CareQ can meaningfully improve the patient experience while simultaneously reducing unnecessary pressure on the healthcare system.

4.5 hours	**60%**	**8 Features**	**5 Members**
Average clinic wait time in Malaysia	Of clinic visits are preventable	Designed to solve real problems	Behind this proposal

2. Introduction & Problem Statement

2.1 Background

Access to healthcare in Malaysia — both through government clinics (Klinik Kesihatan) and private medical facilities — is frequently disrupted by a set of systemic issues that have persisted for years. Long queues, unpredictable waiting times, and a lack of advance appointment systems have become normalised experiences for patients across the country.

These inefficiencies are not simply inconveniences. They represent a fundamental breakdown in the relationship between patients and the healthcare system. Patients are forced to take time off work, travel to clinics without knowing how busy they will be, wait hours for consultations that may last only minutes, and leave without a clear record of what was discussed or prescribed.

On the provider side, clinics face equally pressing challenges. Walk-in volumes are unpredictable, administrative burdens are heavy, and staff must constantly manage crowds rather than focusing on patient care. The result is a system under pressure from both ends.

2.2 The Core Problems

Long Queues & Waiting Times

Government clinics in Malaysia regularly see patients waiting 3 to 5 hours for a single consultation. There is no mechanism for patients to check how busy a clinic is before visiting, nor any way to reserve a time slot. Walk-in queues are the default, creating bottlenecks that affect both patients and clinic operations.

No Appointment System

The majority of government health facilities do not offer any form of advance appointment booking for general consultations. This forces patients into a walk-in system regardless of the urgency or simplicity of their condition, creating unnecessary congestion.

Lack of Cost Transparency

Patients visiting private clinics frequently have no idea how much a consultation will cost until after the appointment. This creates financial anxiety, particularly for lower-income patients, and can discourage people from seeking necessary care.

Absence of Self-Monitoring Tools

Most Malaysians have no structured way to monitor basic health metrics at home. Blood pressure, blood sugar, heart rate — these are only checked reactively, when something has already gone wrong. Early detection and preventive care remain underutilised.

Fragmented Health Records

Patients carry no consolidated health record from clinic to clinic. Doctors are often working blind, unaware of past diagnoses, ongoing medications, or previous test results, leading to repeated tests, slower consultations, and potential medication conflicts.

3. Objectives

CareQ is designed with a clear set of objectives that address each of the core problems identified in the previous section. These objectives guide every feature and design decision within the application.

1. Reduce Patient Waiting Times

By enabling advance appointment booking and real-time queue visibility, CareQ aims to significantly reduce the amount of time patients spend waiting at clinics. Patients can plan their visits, arrive at the right time, and avoid unnecessary congestion.

2. Empower Patients with Self-Monitoring Tools

CareQ provides individuals with the tools to monitor their own basic health metrics — blood pressure, heart rate, blood glucose, and body temperature — and to track these readings over time. This promotes proactive health management and early detection of potential issues.

3. Improve Cost Transparency

By surfacing estimated consultation fees and potential test costs before a visit, CareQ reduces financial uncertainty for patients, especially those attending private facilities. Patients can make informed decisions about where to seek care.

4. Facilitate Better Clinical Consultations

A digital health record stored on the app allows patients to share their medical history, ongoing medications, and past diagnoses with their doctor in seconds. This leads to more informed, efficient, and accurate consultations.

5. Promote Preventive Healthcare

Lifestyle guidance tools — including calorie recommendations, meal planning suggestions, and exercise targets — encourage users to build healthier habits. The goal is to shift the culture from reactive healthcare to proactive wellness.

6. Reduce Unnecessary Clinic Visits

The AI-assisted symptom checker helps users make confident decisions about whether to seek medical attention or manage symptoms at home, reducing unnecessary visits and helping to ease pressure on the healthcare system.

4. Proposed Solution - CareQ

4.1 What is CareQ?

CareQ is a mobile application designed to serve as a complete healthcare companion for Malaysians. The name reflects its core purpose: making care accessible on your queue — on your terms, at your time. It is not simply a digital queue system. It is a unified platform that addresses the full lifecycle of a patient's healthcare journey, from the moment they notice a symptom to their ongoing health management at home.

CareQ is designed to be accessible to all Malaysians regardless of whether they use government or private healthcare facilities. It works across both sectors, providing a consistent experience for users regardless of where they choose to seek care.

4.2 Design Philosophy

Patient-First	Every feature is designed around what the patient needs, not what is convenient for the system.
Simple & Accessible	The app is designed to be usable by all age groups, with a clean interface that requires no technical knowledge.
Preventive, Not Reactive	CareQ encourages users to manage their health proactively, before problems escalate.
Transparent	From queue times to consultation costs, CareQ surfaces information that patients deserve to know upfront.

5. Core Features

CareQ is built around eight core features, each addressing a specific gap in the current patient experience. Together they form a complete healthcare companion that works before, during, and after a clinic visit.

5.1 Appointment Booking

The appointment booking feature allows patients to reserve timeslots at both government and private clinics in advance. Users can browse available slots, select a convenient time, and receive a confirmation — eliminating the need for walk-in queuing entirely.

- Available for both government (Klinik Kesihatan) and private clinics
- Browse available slots by date and time
- Receive booking confirmation and reminders
- Reschedule or cancel appointments easily
- Reduces walk-in congestion and wait times significantly

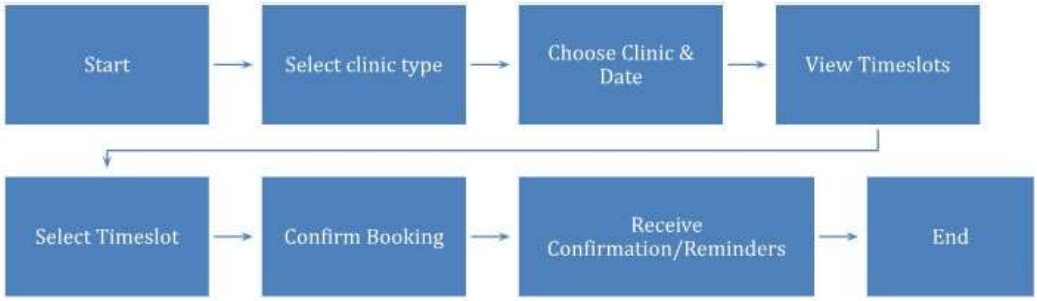
Module 1 Workflow: Appointment Booking Process

The following workflow describes the step-by-step process a user follows when booking an appointment through CareQ.

Step	Description
Step 1	User starts the appointment booking process.
Step 2	User selects clinic type — Government (Klinik Kesihatan) or Private.
Step 3	User chooses a specific clinic and selects a preferred date.
Step 4	System displays available timeslots for the chosen clinic and date.
Step 5	User selects a preferred timeslot.
Step 6	User reviews and confirms the booking.
Step 7	System sends a booking confirmation and sets up reminders.
Step 8	Process ends — appointment is secured.

Process Flowchart — Appointment Booking:

Process Flowchart:



5.2 Real-Time Queue Tracker

Even for patients who cannot book in advance, the real-time queue tracker provides live visibility into clinic congestion. Before leaving home, a patient can check exactly how many people are ahead and receive an estimated wait time.

- Live patient count and estimated waiting time
- Queue updates in real time as patients are seen
- Notifications when your turn is approaching
- Helps patients plan their journey and arrival time
- Reduces anxiety and uncertainty of walk-in visits

Module 2 Workflow: Real-Time Queue Tracker Process

The following workflow describes how the real-time queue tracker operates for walk-in patients.

Step	Description
Step 1	User starts the queue tracking process.
Step 2	User selects the clinic they intend to visit.
Step 3	System fetches live queue data from the selected clinic.
Step 4	App displays the current patient count and estimated wait time.
Step 5	User reviews queue information and decides when to leave for the clinic.
Step 6	User receives a notification as their turn approaches.
Step 7	Process ends — user arrives informed and on time.

5.3 Health Metrics Log

The health metrics log allows users to manually enter their vital readings — blood pressure, heart rate, body temperature, and blood glucose levels — and track these over time. The app visualises trends and flags readings outside healthy ranges.

- Log blood pressure, heart rate, glucose, and temperature
- View historical trends through simple charts
- Receive gentle alerts when readings are abnormal
- Share logged data with doctors during consultations
- Particularly useful for patients managing chronic conditions

5.4 AI-Assisted Symptom Checker

The symptom checker is a guided questionnaire powered by an AI decision-support model. It is not designed to diagnose — it is designed to help patients make confident, informed decisions about whether to seek medical attention or manage their symptoms at home.

- Step-by-step guided symptom questionnaire
- Covers common complaints such as fever, cough, headache, and fatigue
- Recommends home care, monitoring, or clinic visit based on responses
- Reduces unnecessary clinic visits for mild conditions
- Clearly communicated as a support tool, not a diagnostic service

5.5 Clinic Locator

The clinic locator uses the patient's current location to identify nearby clinics and presents them with key information to help choose the most suitable facility.

- Map-based view of nearby government and private clinics
- Filter by opening hours, services offered, and current crowd level
- View estimated consultation costs before visiting
- One-tap navigation via integrated maps
- Helps patients find the right clinic, not just the nearest one

5.6 Estimated Cost Display

Financial uncertainty is a significant barrier to healthcare access in Malaysia. The estimated cost feature provides patients with a transparent view of likely consultation fees and test costs before they commit to a visit.

- Displays estimated consultation fees by clinic
- Shows likely costs for common tests and procedures
- Helps patients compare costs across facilities
- Particularly valuable for uninsured or cost-sensitive patients
- Promotes financial planning and informed decision-making

5.7 Digital Health Record

CareQ serves as a portable digital health record, allowing users to store and carry their medical history at all times. This record is available to share with any doctor, regardless of which clinic the patient visits.

- Store past diagnoses, medications, and vaccination history
- Keep records of test results and lab reports
- Share records with doctors via a secure in-app sharing feature
- Medication reminders and refill alerts
- Reduces repeated tests and improves consultation quality

5.8 Lifestyle Guidance

Recognising that health extends far beyond clinic visits, CareQ provides personalised lifestyle guidance to help users build healthier daily habits. These tools are designed to be gentle nudges, not overwhelming targets.

- Personalised daily calorie recommendations based on age, weight, height, and activity
- Simple balanced meal suggestions and hydration reminders
- Exercise recommendations aligned with WHO guidelines (150 min/week)
- Daily health tips and wellness content
- Encourages long-term habit formation, not short-term fixes

6. User Journey & Experience

The following describes a typical end-to-end user journey through CareQ, illustrating how the features work together as a seamless experience from feeling unwell at home to post-visit health management.

Step 1: Symptom Onset at Home

The user notices they are feeling unwell — perhaps a persistent headache, mild fever, or unusual fatigue. Rather than immediately heading to a clinic, they open CareQ and use the symptom checker. After answering a short guided questionnaire, the app advises whether to rest at home, monitor symptoms over the next 24 hours, or seek medical attention promptly.

Step 2: Finding the Right Clinic

If a clinic visit is recommended, the user opens the clinic locator. They can see nearby government and private clinics on a map, view current queue levels, check opening hours, see the services offered, and view estimated consultation costs — all before leaving home.

Step 3: Booking a Timeslot

The user selects a clinic and books an available timeslot that fits their schedule. They receive a confirmation notification and a reminder closer to the time. There is no need to arrive early and wait in a physical queue.

Step 4: Arriving at the Clinic

The user arrives at their scheduled time. If they are a walk-in patient, they can check the real-time queue tracker to monitor their position and receive a notification when their turn is near. The experience is predictable rather than anxious.

Step 5: The Consultation

Inside the clinic, the user can share their digital health record with the doctor through the app. The doctor has immediate access to the patient's past diagnoses, ongoing medications, vaccination history, and recent health readings — making the consultation faster, more informed, and more effective.

Step 6: Post-Visit & Ongoing Monitoring

After the visit, the user continues to log their health readings, take their medications on schedule (aided by app reminders), and track their recovery. The lifestyle guidance tools support them in building healthier daily habits that reduce the likelihood of future clinic visits.

7. Technical Overview

7.1 Platform & Architecture

CareQ is proposed as a cross-platform mobile application, available on both iOS and Android devices. The architecture follows a client-server model with a cloud-hosted backend to ensure scalability and real-time data synchronisation across users and clinics.

Component	Proposed Technology	Purpose
Mobile App	React Native / Flutter	Cross-platform iOS & Android
Backend API	Node.js / Python (FastAPI)	Core business logic & data handling
Database	PostgreSQL + Redis	Patient data & real-time queue cache
Authentication	JWT + OAuth 2.0	Secure user login & session management
Cloud Hosting	AWS / Google Cloud	Scalable infrastructure & uptime
AI Symptom Engine	Python ML Model	Symptom decision support logic
Maps Integration	Google Maps API	Clinic locator & navigation
Push Notifications	Firebase Cloud Messaging	Queue alerts & reminders

7.2 Data Privacy & Security

Health data is among the most sensitive categories of personal information. CareQ is designed with privacy and security as foundational requirements, not afterthoughts. All health records are stored with end-to-end encryption. User consent is required before any data is shared. The application will comply with Malaysia's Personal Data Protection Act (PDPA) and any applicable digital health regulations.

7.3 Clinic Integration

For CareQ to function at full capacity, integration with clinic management systems is required. The proposal envisions a phased integration model — beginning with clinics that use digital management tools and expanding to others over time. An API-based integration layer will allow CareQ to sync appointment slots, queue data, and basic service information from participating clinics.

8. Expected Impact & Benefits

8.1 Benefits for Patients

- Significantly reduced time spent waiting at clinics
- Greater confidence and control in making healthcare decisions
- Early detection of health issues through ongoing self-monitoring
- Financial clarity before attending a private clinic
- A portable, comprehensive health record always within reach
- Reduced need for unnecessary clinic visits through the symptom checker
- Better long-term health outcomes through lifestyle guidance

8.2 Benefits for Clinics

- More predictable patient flow through advance booking
- Reduced walk-in congestion and administrative burden
- Better-informed patients arriving with their health history available
- Fewer unnecessary consultations for minor or self-manageable conditions
- Improved patient satisfaction and clinic reputation

8.3 Benefits for the Healthcare System

- Reduced overall pressure on government clinic resources
- Promotion of preventive healthcare at scale
- Greater digital health infrastructure across Malaysia
- Data insights to support healthcare planning and resource allocation
- A foundation for future expansion into telemedicine and digital health services

9. Target Users

CareQ is designed for all Malaysians who interact with the healthcare system. However, the following user segments represent the primary target audience for the application.

General Public (Ages 18–60)

The primary user group — working adults who need to visit clinics but cannot afford to spend hours waiting. This group benefits most from appointment booking, queue tracking, and cost transparency.

Elderly Patients & Caregivers

Older Malaysians managing chronic conditions such as hypertension and diabetes benefit significantly from the health metrics log, medication reminders, and digital health record features. Caregivers managing appointments on behalf of elderly relatives are also a key segment.

Chronic Condition Patients

Patients with ongoing conditions — diabetes, hypertension, asthma, and similar — who require regular monitoring and frequent clinic visits. The health metrics log and digital record are particularly valuable for this group.

Health-Conscious Individuals

Users who are proactively managing their wellness through exercise, nutrition, and preventive monitoring. The lifestyle guidance and metrics tracking features serve this segment.

Rural & Underserved Communities

Patients in areas with limited healthcare access who benefit from the clinic locator and symptom checker — reducing the need to travel to a clinic for conditions that can be managed at home.

10. Implementation Roadmap

CareQ is proposed for development in three distinct phases, each building on the previous. This phased approach allows for controlled testing, feedback integration, and progressive scaling without overcommitting resources in the early stages.

Phase	Timeline	Deliverables
Phase 1: Foundation & Core Features	Months 1–4	Appointment booking, real-time queue tracker, clinic locator, user accounts, pilot with 10–20 clinics in the Klang Valley.
Phase 2: Health Monitoring & Records	Months 5–9	Health metrics logging, digital health record, medication reminders, AI symptom checker (v1), estimated cost display for all clinic partners.
Phase 3: Lifestyle & Expansion	Months 10–15	Lifestyle guidance module, expanded AI symptom checker, national rollout, insurance partnerships, analytics dashboard for clinic partners.

11. Challenges & Mitigation

Every technology initiative faces challenges. The following table identifies the key risks associated with CareQ and the proposed mitigation strategies for each.

Challenge	Description	Mitigation Strategy
Clinic Adoption	Government and private clinics may be slow to integrate with the platform.	Begin with tech-ready private clinics; work with KKM for government rollout.
Digital Literacy	Older users may struggle with app-based tools.	Simple, clear UI design; tutorial walkthrough on first launch; support hotline.
Data Privacy	Patients may be hesitant to store health records digitally.	Full PDPA compliance; transparent privacy policy; opt-in data sharing only.
AI Accuracy	The symptom checker may give inaccurate guidance.	Clear disclaimer that it is not a diagnostic tool; conservative recommendations.
Internet Connectivity	Rural users may have limited data access.	Offline mode for core features; lightweight app design; local caching.
Scaling Infrastructure	High user volumes may strain backend systems.	Cloud-hosted infrastructure with auto-scaling; load testing before launch.

12. Conclusion

The challenges facing the Malaysian healthcare system are real, persistent, and felt by millions of people every day. Long queues, opaque costs, fragmented records, and a culture of reactive rather than preventive healthcare are problems that digital solutions are uniquely positioned to address.

CareQ is proposed as a meaningful step toward a more patient-centred healthcare experience. By giving patients the tools to book appointments, track their health, make informed decisions, and carry their medical history with them, the application addresses root causes rather than symptoms.

This proposal does not claim that CareQ will solve all of Malaysia's healthcare challenges. It does claim that the right digital tools, designed thoughtfully and deployed carefully, can make the system meaningfully better for patients, clinics, and the broader healthcare infrastructure.

As Malaysia continues its journey toward a digital-first healthcare ecosystem, CareQ represents a practical, achievable, and patient-focused contribution to that future. The team behind this proposal is confident in both the problem identified and the solution proposed, and welcomes the opportunity to develop this concept further.

CareQ

Healthcare, finally on your schedule.

Prepared by: Kasem Bau · Kemi Yonaniko · Hardev Singh · Nor Shahira · Noraisyah